

From MOOC to MAIC: Reshaping Online Teaching and Learning through LLM-driven Agents

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Abstract

Since the first instances of online education, where courses were uploaded to accessible and shared online platforms, this form of scaling the dissemination of human knowledge to reach a broader audience has sparked extensive discussion and widespread adoption. Recognizing that personalized learning still holds significant potential for improvement, new AI technologies have been continuously integrated into this learning format, resulting in a variety of educational AI applications such as educational recommendation and intelligent tutoring. The emergence of intelligence in large language models (LLMs) has allowed for these educational enhancements to be built upon a unified foundational model, enabling deeper integration. In this context, we propose MAIC (Massive AI-empowered Course), a new form of online education that leverages LLM-driven multi-agent systems to construct an AI-augmented classroom, balancing scalability with adaptivity. Beyond exploring the conceptual framework and technical innovations, we conduct preliminary experiments at Tsinghua University, one of China’s leading universities. Drawing from over 100,000 learning records of more than 500 students², we obtain a series of valuable observations and initial analyses. This project will continue to evolve, ultimately aiming to establish a comprehensive open platform that supports and unifies research, technology, and applications in exploring the possibilities of online education in the era of large model AI. We envision this platform as a collaborative hub, bringing together educators, researchers, and innovators to collectively explore the future of AI-driven online education.

1 Introduction

Explicit Background: Evolution for Scalability. The evolution of online education stands as a testament to humanity’s relentless pursuit of knowledge, transcending the limitations of time and space (Koller and Ng, 2013). From the humble beginnings of oral tradition to the advent of the printed book (Froebel, 1886; Halstead and Taylor, 2005), education has continually sought ways to expand its reach. Yet, for centuries, the traditional model of education was bound by the constraints of physical classrooms, limited resources, and localized instruction. The dawn of the internet marked

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从慕课到 MAIC：通过LLM驱动的智能体重塑在线教学与学习

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摘要

自在线教育最初将课程上传至开放共享的网络平台以来，这种扩大人类知识传播范围以覆盖更广泛受众的方式引发了广泛讨论并获得普遍采用。鉴于个性化学习仍具有显著改进潜力，新型人工智能技术持续融入该教育模式，催生了教育推荐系统、智能辅导系统等各类教育AI应用。大型语言模型（LLM）中智能技术的涌现，使得这些教育升级方案得以基于统一基础模型构建，实现深度整合。在此背景下，我们提出 MAIC（大规模AI赋能课程）这一新型在线教育模式，通过LLM驱动的多智能体系统构建AI增强型课堂，实现可扩展性与适应性的平衡。

除探索概念框架与技术创新外，我们还在中国顶尖学府清华大学开展了初步实验。

我们基于超过500名学生逾10万条学习记录²，获得了系列有价值的观察结果与初步分析。该项目将持续发展完善，最终目标是构建一个综合性开放平台，通过整合研究、技术与应用资源，探索大型模型人工智能时代在线教育的发展潜力。我们构想该平台将成为协作枢纽，汇聚教育工作者、科研人员与创新者，共同探索人工智能驱动型在线教育的未来发展方向。

1 引言

明确背景：为**可扩展性**而演进。在线教育的发展历程见证了人类对知识不懈追求的精神，突破了时空限制（Koller 与 Ng, 2013）。从口述传统的朴素起源到印刷书籍的出现（Froebel, 1886; Halstead 与 Taylor, 2005），教育始终在探索扩大其覆盖范围的方式。然而数百年来，传统教育模式受限于实体教室、资源匮乏及地域化教学的桎梏。互联网时代的到来标志着

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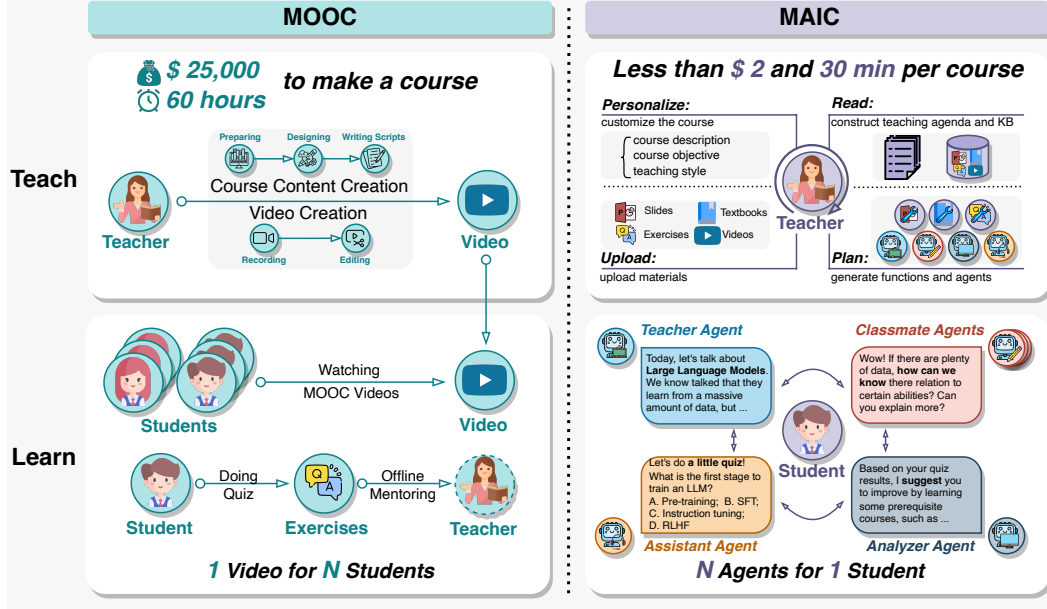


Figure 1: MOOC V.S. MAIC from the aspects of teaching and learning.

a revolutionary shift, heralding the age of online education, where the dream of universal access to knowledge began to take tangible form. Specifically, the Massive Open Online Course (MOOC) phenomenon marks a significant milestone in the evolution of online education, reflecting both technological advancement and educational innovation (Cormier, 2008; Daniel, 2012). Since then, platforms like edX³, involving institutions such as MIT and Harvard, and Coursera⁴, originating from Stanford, have integrated learning resources from over 270 renowned universities (Pappano, 2012; Papadakis, 2023). These platforms have attracted more than 100 million learners globally, progressively realizing the *Scalability* of online education.

Implicit Motivation: Determination of Adaptivity. However, this paradigm of serving thousands of learners from diverse backgrounds through one pre-recorded video (Reich and Ruipérez-Valiente, 2019) (as shown in Figure 1) struggles to align with the educational philosophy of “teaching in accordance with individual aptitudes” (Reich, 2015; Zhu et al., 2020). This challenge has become a significant reason for the subsequent efforts that introducing of AI techniques into online learning. To achieve the *Adaptivity* of learning, a series of tasks such as learning path planning (Nabizadeh et al., 2020; Zhong et al., 2022), course recommendation (Jiang et al., 2019b; Zhang et al., 2018; Jing and Tang, 2017), and intelligent tutoring (Yilmaz et al., 2022; Tu et al., 2023)—driven by technologies like recommendation systems and dialogue generation—have been employed to enhance the student learning experience.

Although these technologies have been applied across various aspects of teaching and learning, the significant differences among the supporting tasks before, during, and after instruction have posed challenges for unifying them under a single deep learning framework (Kabudi et al., 2021). Such fragmentation has, in part, delayed the emergence of a new platform where AI and online learning are fully integrated. However, with the rapid advancement of generative AI (Epstein et al., 2023), large language models (LLMs) have created fresh opportunities for AI-powered learning paradigms. Models such as GPT-4 (Achiam et al., 2023) and LLaMA (Touvron et al., 2023) possess strong generalization capabilities and encapsulate vast parametric knowledge, allowing for the flexible configuration of intelligent agents (Chen et al., 2023) built upon them. Currently, LLM-driven multi-agent systems (Wu et al., 2023) have already been explored for applications such as social simulation (Park et al., 2023a) and the execution of complex tasks like software development (Qian et al., 2023). This progress opens up a potential pathway for introducing large language model multi-agent systems to create entirely new online teaching and learning experiences.

³<https://www.edx.org/>

⁴<https://www.coursera.org/>

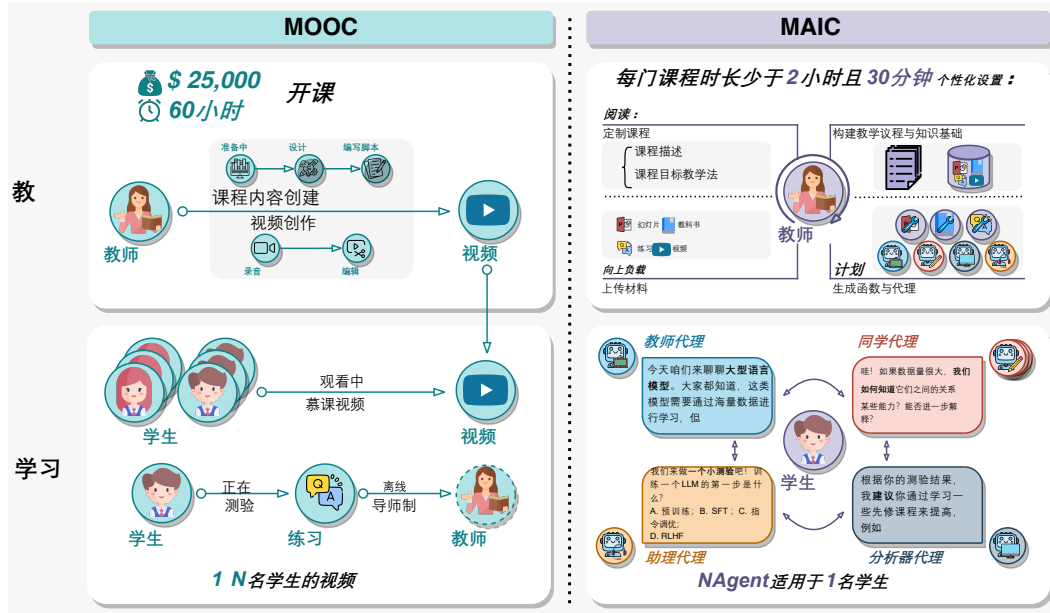


图1：从教学与学习角度比较MOOC与 MAIC

一场革命性的转变，预示着在线教育时代的到来，普及知识的梦想开始具象化。具体而言，大规模开放在线课程（MOOC）现象标志着在线教育发展的重要里程碑，既反映了技术进步也体现了教育创新（Cormier, 2008; Daniel, 2012）。此后，edX³（涉及麻省理工学院、哈佛大学等机构）与Coursera⁴（源自斯坦福大学）等平台整合了270余所知名高校的学习资源（Pappano, 2012; Papadakis, 2023）。这些平台已吸引全球逾1亿学习者，逐步实现在线教育的**可扩展性**。

内隐动机：适应性测定。然而，通过一段预录视频为数千名不同背景的学习者提供服务的范式（Reich 与 Ruiperez-Valiente, 2019）（如图1所示）难以与“根据个体能力进行教学”的教育理念相契合（Reich, 2015; Zhu 等人, 2020）。这一挑战已成为后续将人工智能技术引入在线学习的重要动因。为实现学习的**适应性**，一系列任务被提出，包括学习路径规划（Nabizadeh等人, 2020; Zhong等人, 2022）、课程推荐（Jiang等人, 2019b; Zhang等人, 2018; Jing与Tang, 2017）以及智能辅导（Yilmaz 等人, 2022; Tu 等人, 2023）——这些任务由技术驱动推荐系统与对话生成技术已被应用于提升学生学习体验。

尽管这些技术已应用于教学与学习的各个领域，但教学前、教学中及教学后支持任务之间的显著差异，给将其统一于单一深度学习框架内带来了挑战（Kabudi 等人, 2021）。这种碎片化现象在一定程度上延缓了人工智能与在线学习深度融合新平台的出现。然而，随着生成式人工智能的快速发展（Epstein等人, 2023），大型语言模型（LLMs）为人工智能驱动的学习范式创造了全新机遇。诸如GPT-4（Achiam 等人, 2023）和LLaMA（Touvron 等人, 2023）等模型具备强大的泛化能力，并封装了海量参数化知识，使得基于其构建的智能体（Chen 等人, 2023）能够灵活配置。目前，LLM驱动的多智能体系统（Wu 等人, 2023）已在社交模拟等应用场景中得到探索。Park等人（2023a）以及执行软件开发等复杂任务（Qian等人, 2023）。这一进展为引入大型语言模型多智能体系统开辟了潜在途径，从而创造出全新的在线教学与学习体验。

³<https://www.edx.org/>

⁴<https://www.coursera.org/>

Proposal of MAIC. At the critical juncture of a new era defined by large language models and multi-agent systems in online education, we introduce MAIC (Massive AI-empowered Course). MAIC is dedicated to exploring the integration of multi-agent systems across various stages of online learning, including course preparation, instruction, and analysis, with the goal of balancing *Scalability* and *Adaptivity* of online education. The core concept of MAIC is to construct a series of LLM-driven agents to support both **Teaching** and **Learning** in the online educational environment. As illustrated in the Figure 1, the paradigm of MOOC and MAIC can be featured with two primary aspects:

Teaching: This action is primarily performed by the instructor. In previous MOOC, the instructor is responsible for thoroughly preparing course materials, drafting lecture notes, and spending considerable time meticulously recording the course. The final output typically consists of a series of pre-recorded instructional videos. For the proposed MAIC, however, the instructor only needs to upload the teaching slides. With intelligent assistance, the instructor can further complete the PPT creation, after which the agents, utilizing a range of models such as multimodal understanding and knowledge structure extraction, will generate structured lecture notes and learning resources optimized for use by the AI system. **Learning:** In MOOCs (Reich and Ruipérez-Valiente, 2019), a single set of course materials is designed to serve thousands of students with diverse backgrounds, and the pace of instruction is predetermined by the instructor, offering very limited room for personalized adaptation based on individual student needs (Yu et al., 2021). While in MAIC, course delivery is autonomously managed by AI teacher agents, which dynamically adjust the teaching process based on student interactions and inquiries. Additionally, MAIC offers AI teaching assistants and customizable AI classmates. Students can select the AI agents they wish to study with, thereby creating varied classroom scenarios that provide personalized learning companions, emotional support, and opportunities for knowledge discussion.

In this report, we introduce the MAIC platform, offering an intuitive and user-friendly solution that accommodates the needs of various users, including students and educators. This platform comes pre-equipped with a suite of intelligent agents and tools that support course analysis and the construction of new MAIC course examples. Additionally, MAIC integrates several learning analytics tools powered by large models, enabling quick access to learning data, forecasting academic outcomes, and automating tasks such as interviews and assessments.

With the support of Tsinghua University, one of China’s top universities, we conducted an exploration of this new learning model over a period of more than three months. Assisted by over 500 student volunteers, we implemented the study using two courses: the AI course “Towards Artificial General Intelligence” (TAGI) and the learning science course “How to Study in the University” (HSU). During this pilot, we collected over 100,000 behavioral records. Based on the data from these courses, along with student survey measurements and qualitative interview results, we conducted an initial analysis of the features and performance of the MAIC system. In subsequent sections, we will briefly introduce the technical implementation, algorithm improvement and primary results of MAIC ⁵.

2 Related Work

AI-assisted Online Learning Online learning (Pappano, 2012) refers to the process of acquiring knowledge within an electronic environment composed of communication technologies, network infrastructure, artificial intelligence, and multimedia tools. While online education significantly enhances learners’ access to knowledge, supports personalized learning, and facilitates learning tailored to individual needs, persistent issues such as low course completion rates and suboptimal learning outcomes remain formidable challenges (Zhong et al., 2022). A retrospective study published by MIT (Reich and Ruipérez-Valiente, 2019) highlights that the lack of continuous guidance and personalized support for online learners is a critical factor affecting the quality of learning and student development in online settings. The inherent nature of online and remote learning often results in physical separation between students and instructors, making it prohibitively expensive to maintain real-time interactions through manual means (Aleven et al., 2015). Therefore, AI researchers have begun to introduce auxiliary learning applications such as resource recommendation systems (Jing and Tang, 2017) and intelligent teaching assistants (Jiang et al., 2019a; Tu et al., 2023) into online education environments. Leveraging technologies like educational knowledge graphs (Dang et al.,

⁵Our open-source Demo and detailed analysis will be released soon. More technical details are introduced in the companion papers Slide2Lecture and SimClass (Zhang et al., 2024).

MAIC 提案。在由大型语言模型和多智能体系统定义的在线教育新时代的关键节点，我们提出 MAIC（大规模AI赋能课程）。MAIC 致力于探索多智能体系统在在线学习各阶段（包括课程准备、教学与分析）的整合应用，旨在平衡在线教育的**可扩展性**与适应性。MAIC 的核心理念是构建一系列由LLM驱动的智能体，以支持在线教育环境中的**教学与学习**。如图1所示，MOOC与 MAIC 的范式可从两个方面体现：

教学：这一环节主要由讲师完成。在以往的慕课中，讲师需要全面准备课程材料、撰写讲义，并花费大量时间精心录制课程。最终输出通常是一系列预先录制的教学视频。然而对于提议的 MAIC，讲师只需上传教学幻灯片即可。在智能辅助下，讲师可进一步完成PPT制作，随后智能体将利用多模态理解、知识结构提取等多种模型，生成针对AI系统优化的结构化讲义和学习资源。**学习：**在慕课（Reich 与 Ruiperez-Valiente, 2019）中，一套课程材料需服务于背景各异的数千名学生，且教学节奏由讲师预先设定，基于个体学生需求进行个性化调整的空间极为有限（Yu 等人, 2021）。在 MAIC 期间，课程教学由AI教师智能体自主管理，这些智能体会根据学生互动与提问动态调整教学流程。此外，MAIC 还提供AI教学助理和可定制的AI同学。学生可自主选择学习伙伴，从而构建多样化的课堂场景，为学生提供个性化学习陪伴、情感支持及知识交流机会。

本报告重点介绍 MAIC 平台，该平台提供直观易用的解决方案，可满足学生、教育工作者等各类用户需求。平台预装智能代理工具套件，支持课程分析及新 MAIC 案例开发。此外，MAIC 整合了基于大型模型的学习分析工具，可快速获取学习数据、预测学业成果，并实现访谈评估等任务的自动化处理。

在清华大学（中国顶尖学府之一）的支持下，我们历时三个月开展了新型学习模式的探索研究。在500余名学生志愿者协助下，通过两门课程实施该研究：人工智能课程《迈向通用人工智能》（TAGI）与学习科学课程《大学学习指南》（HSU）。试点期间共收集超过10万条行为记录数据。基于课程数据、学生问卷调查结果及定性访谈资料，我们对 MAIC 系统的核心特征与运行表现进行了初步分析。后续章节将简要介绍技术实现方案、算法优化方案及 MAIC 主要研究成果⁵。

2 相关工作

人工智能辅助在线学习在线学习（Pappano, 2012）是指在由通信技术、网络基础设施、人工智能和多媒体工具构成的电子环境中获取知识的过程。尽管在线教育显著提升了学习者获取知识的途径、支持个性化学习并促进针对个体需求的学习，但课程完成率低和学习效果欠佳等持续性问题仍是严峻挑战（Zhong等人, 2022）。麻省理工学院发表的一项回顾性研究（Reich 与 Ruiperez-Valiente, 2019）指出，在线学习者缺乏持续指导和个性化支持是影响在线环境中学习质量与学生发展的关键因素。在线远程学习的固有特性常导致师生之间存在物理隔离，使得通过人工方式维持实时互动成本过高（Aleven 等人, 2015）。因此，人工智能研究人员已开始将辅助学习应用引入在线教育环境，例如资源推荐系统（Jing与 Tang, 2017）和智能教学助理（Jiang等人, 2019a; Tu等人, 2023）。通过利用教育知识图谱等技术（Dang等人,

⁵我们的开源演示版本及详细分析报告即将发布。更多技术细节详见配套论文Slide2Lecture与SimClass（Zhang等人, 2024）。

2019; Yu et al., 2020), they are increasingly integrating diverse technologies to construct personalized online learning systems that enhance the learning experience through tailored support and adaptive learning pathways. In the era of large language models, platforms like Khan Academy have pioneered the deployment of AI-driven tools, such as the Khanmigo virtual tutor ⁶. This development has sparked discussions among researchers about the potential to move beyond basic question-answering and recommendation functionalities, exploring the design of more deeply integrated models that fuse AI with online education in innovative ways (Golchin et al., 2024). Some of the research efforts gradually aim to create new paradigms that go beyond traditional approaches to AI-enhanced learning.

LLM-driven AI Tutoring System The evolution of intelligent tutoring systems (ITS) from early expert systems to agent-based models in the era of large language models has undergone several transformations in interaction paradigms (Pal Chowdhury et al., 2024). During the 1980s and 1990s, ITS began leveraging expert systems and related technologies to deliver instruction aligned with learners’ cognitive processes across diverse educational settings. Representative systems from this period include AutoTutor, developed by the University of Memphis (Nye et al., 2014), and SCOT from Stanford University[51]. These systems offered greater flexibility and were among the first to support natural language-based question-and-answer interactions. However, the content of these interactions remained pre-designed, limiting their ability to provide learning support beyond the scope of the system’s initial design. The advent of Large Language Models (LLMs) has significantly broadened the scope of intelligent tutoring systems (ITS), offering unprecedented interactivity and adaptability across educational platforms (Bubeck et al., 2023). Recent strides in the study of multi-agent systems and the integration of tools have catalyzed new approaches to planning and student interaction within ITS frameworks (Park et al., 2023b; Qian et al., 2023; Schick et al., 2024). Emerging research (Chen et al., 2024) illustrates the capacity of LLMs to autonomously curate and deliver educational content by leveraging an array of tools. Additionally, innovations like MWPTutor (Pal Chowdhury et al., 2024) explore how LLMs can effectively manage the teaching of complex subjects, such as mathematical word problems, demonstrating their versatility in specialized learning environments. Building on these developments, our research focuses on an LLM-based ITS model designed to establish a robust framework for comprehensive, lecture-level tutoring that aligns with evolving educational standards, thereby reinforcing the transformative role of LLMs in the future of education.

3 MAIC

In this section, we introduce the key techniques for implementing a MAIC platform. Specifically, we present the main workflows designed for both the teaching and learning sides, highlighting the key challenges associated with each and the corresponding solutions implemented to address these issues.

3.1 Teaching: Course Preparation Workflow

To transform vast amounts of weakly structured and static learning resources (Dang et al., 2019) into highly structured and adaptive learning materials (Yu et al., 2021), MAIC develop a standardized course preparation workflow. This workflow is designed to streamline the workload of experts, facilitating the scalability of this online learning model and preparing it for broader implementation. Such course preparation workflow of MAIC consists of two major stage: **Read** and **Plan**.

3.1.1 Read Stage

At this stage, instructors (and authorized teaching assistants) are involved by providing material. With the assistance of multi-agent systems empowered by large language models, they upload a set of course slides $\mathcal{P} = \{P_i\}_{1 \leq i \leq |\mathcal{P}|}$, which are then transformed into highly structured intelligent learning resources $\hat{\mathcal{P}} = \{\langle P_i, D_i, K_j \rangle\}_{1 \leq i \leq |\mathcal{P}|, 1 \leq j \leq |\mathcal{K}|}$ along with multiple AI agents designed for classroom construction. The P_i and D_i here correspond to a single slide page and its textual description, while K_j denotes to the knowledge-aware section of each page.

1. Slides Content Extraction. First, MAIC employs a multi-modal LLM (mLLM) (Huang et al., 2023) to capture the textual content P_i^t and the visual content P_i^v of each page of the given slide $\mathcal{P}$,

⁶<https://blog.khanacademy.org/teacher-khanmigo/>

2019; Yu等人, 2020), 他们正日益整合多样化技术来构建个性化在线学习系统, 通过定制化支持和自适应学习路径提升学习体验。在大型语言模型时代, Khan Academy等平台率先部署了AI驱动工具, 例如Khanmigo虚拟导师⁶。这一发展引发了研究者关于突破基础问答与推荐功能潜力的讨论, 探索以创新方式融合AI与在线教育的深度集成模型设计 (Golchin等人, 2024)。部分研究努力正逐步致力于构建超越传统AI增强学习方法的新范式。

LLM驱动的人工智能辅导系统在大型语言模型时代, 智能辅导系统 (ITS) 从早期专家系统到基于代理模型的演变过程中, 交互范式经历了多次转型 (Pal Chowdhury等人, 2024)。20世纪80至90年代, ITS开始利用专家系统及相关技术, 在多样化教育场景中提供与学习者认知过程相匹配的教学。该时期的代表性系统包括孟菲斯大学开发的AutoTutor (Nye等人, 2014) 以及斯坦福大学的SCOT[51]。这些系统具有更高的灵活性, 并率先支持基于自然语言的问答交互。然而, 这些交互内容仍保持预设形式, 导致其学习支持能力受限于系统初始设计范围。大型语言模型 (LLMs) 的出现显著拓宽了智能辅导系统 (ITS) 的应用范围, 为不同教育平台提供了前所未有的交互性和适应性 (Bubeck 等人, 2023)。多智能体系统研究及工具整合的最新进展, 推动了ITS框架内教学规划与学生互动的新方法 (Park等人, 2023b; Qian等人, 2023; Schick等人, 2024)。新兴研究 (Chen 等人, 2024) 展示了LLMs通过整合多种工具自主策划与传递教育内容的能力。此外, MWPTutor等创新方案 (Pal Chowdhury等人, 2024) 探索了LLMs如何有效管理数学应用题等复杂学科的教学, 彰显了其在专业学习环境中的多功能性。基于这些研究成果, 我们的研究聚焦于一种基于大语言模型 (LLM) 的智能教学系统 (ITS) 模型, 旨在构建一套稳健框架, 实现与不断演变的教育标准相契合的全面性、课时级辅导功能, 从而强化大语言模型在未来教育中的变革性作用。

3 MAIC

本节将介绍实现 MAIC 平台的核心技术。具体而言, 我们将阐述针对教学与学习双方设计的主要工作流程, 重点分析各自面临的关键挑战, 并说明为解决这些问题所采取的相应解决方案。

3.1 教学: 课程准备工作流程

为了将大量结构松散且静态的学习资源 (Dang等人, 2019) 转化为高度结构化且自适应的学习材料 (Yu 等人, 2021), MAIC 开发了一套标准化课程准备工作流程。该流程旨在简化专家的工作负担, 提升这种在线学习模式的可扩展性, 并为其更广泛的应用做好准备。MAIC 课程准备工作流程包含两个主要阶段: **阅读与规划**。

3.1.1 读取阶段

在此阶段, 教师 (及授权助教) 通过提供教学材料参与其中。借助由大型语言模型赋能的多智能体系统, 他们上传一组课程幻灯片 $P = \{P_i\}_{1 \leq i \leq |P|}$, 随后这些幻灯片被转化为高度结构化的智能学习资源 $P = \{P_i, D_i, K_j\}_{1 \leq i \leq |P|, 1 \leq j \leq |PP|}$ 以及多个专为教室建设设计的AI代理。此处的 P_i 和 D_i 对应单个幻灯片页面及其文本描述, 而 K_j 表示每页的知识感知部分。

1. 幻灯片内容提取。首先, MAIC 采用多模态大语言模型 (mLLM) (Huang等人, 2023) 用于捕获给定幻灯片 P 中每页的文本内容 P_i^t 和视觉内容 P_i^v ⁶<https://blog.khanacademy.org/teacher-khanmigo/>

[khanacademy.org/teacher-khanmigo/](https://blog.khanacademy.org/teacher-khanmigo/)

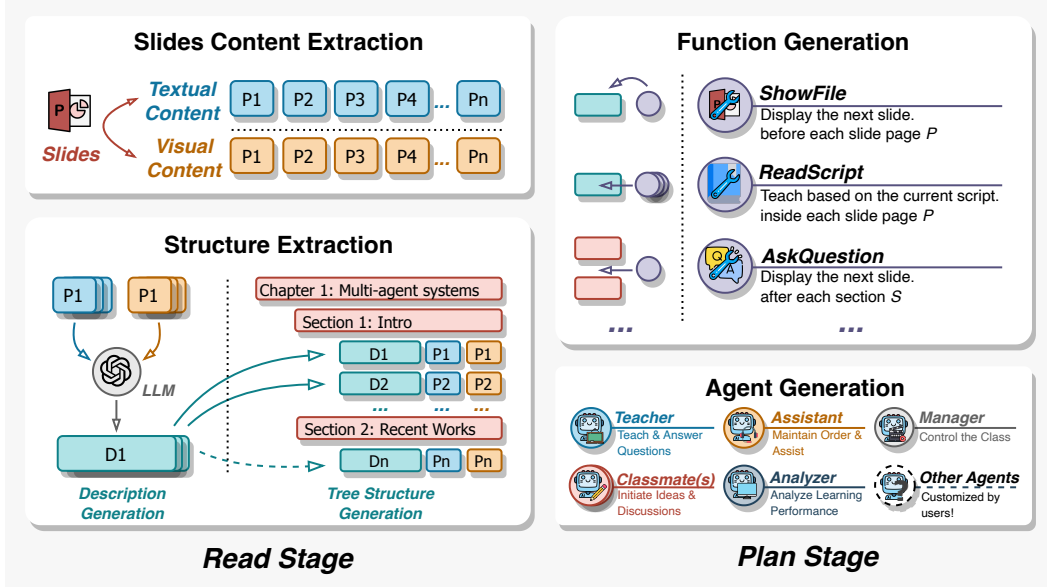


Figure 2: An illustration of the course preparation workflow of MAIC.

i.e., $f_T^1 : P_i \rightarrow \langle P_i^t, P_i^v \rangle$. Such functional model f_T^1 ⁷ can be adjusted and further improved by the arising LLM techniques, and the current implementation is based on the GPT-4V⁸ model with certain prompting contexts.

2. Structure Extraction. After the pre-processing of the uploaded slides, MAIC employs two functions to complete the read stage. The produced $\langle P_i^t, P_i^v \rangle$ are described by an mLLM-based method with straightforward and comprehensive texts, i.e., $f_T^2 : \langle P_i^t, P_i^v \rangle \rightarrow D_i$. Meanwhile, MAIC takes an knowledge extraction method to organize the core knowledge of each page and build a tree-style taxonomy for the slide, i.e., $f_T^3 : \langle P_i^t, P_i^v, D_i \rangle \rightarrow K_j$, which makes up the final $\hat{P}$.

3.1.2 Plan Stage

At this stage, instructors (and authorized teaching assistants) are involved by proofreading and refining the results. Based on highly structured slides, MAIC constructs a novel instructional action representation language, allowing the incorporation of flexible teaching functions such as lecturing and questioning into preset classrooms, which naturally connects with related educational technologies like lecture script generation (Hong et al., 2023b) and question generation (Kurdi et al., 2020). Meanwhile, leveraging intelligent agent construction techniques (Chen et al., 2023), the platform utilizes course content to provide teachers and teaching assistants with AI-driven agents, facilitating the early planning of the foundational structure for subsequent courses.

3. Function Generation. To make the heterogeneous teaching actions be generated within the classroom context, teaching activities such as lecturing and giving quizzes are conceptualized as *teaching actions* within MAIC. Each teaching action $\mathcal{T}$ is defined as $\mathcal{T} = (type, value)$, where *type* indicates the category of the action (e.g., *ShowFile*, *ReadScript*, *AskQuestion*), and *value* details the content of the action, such as the script to be read aloud. This approach reflects our principles of flexibility and adaptability, allowing classroom actions to be easily configurable. It empowers developers and educators to create custom teaching actions tailored to specific needs, facilitating their smooth integration into the overall teaching process.

Each Function is associated with the generation of certain content, denoted as $\langle \mathcal{T}_n, \hat{P}_{\mathcal{T}} \rangle$. For instance, the function *AskQuestion* necessitates linkage with one or a set of specific questions. Among these, the most crucial action is *ReadScript*, as it constitutes the core of the instructional process. Based on this function, these Teaching Actions are embedded within the course script

⁷The same applies to similar symbols in the following text.

⁸<https://openai.com/index/gpt-4v-system-card/>

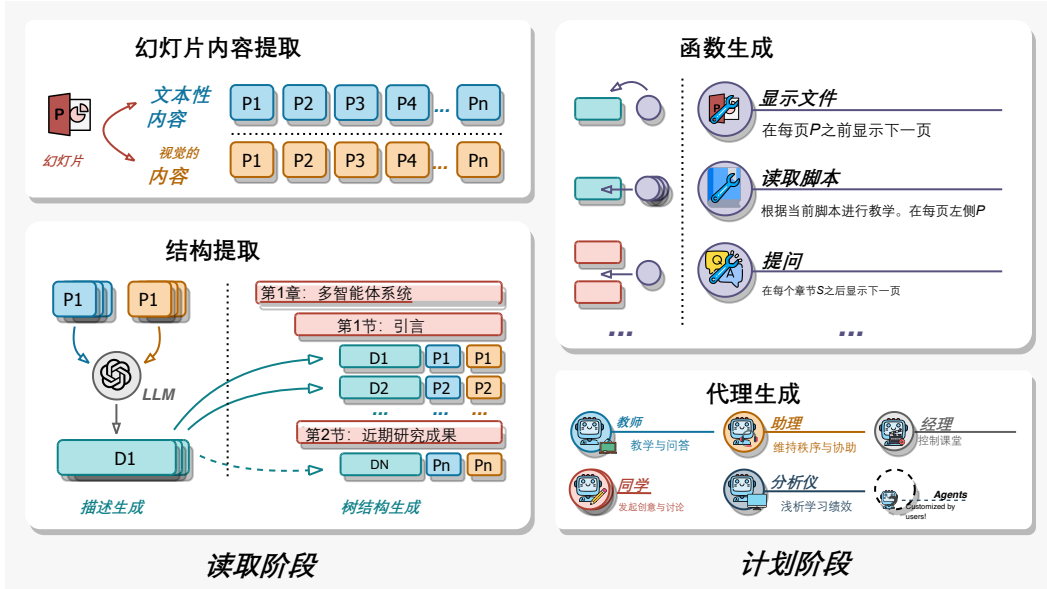


图2：MAIC 课程准备工作流程示意图。

即， $f1T: P_i \rightarrow \langle P_i^t, P_i^v \rangle$ 。此类功能模型 $f1T$ ⁷可通过新兴的LLM技术进行调整和进一步优化，当前实现基于带有特定提示上下文的GPT-4V⁸模型。

2. 结构提取。在上传幻灯片完成预处理后，MAIC 采用两种功能来完成读取阶段。生成的 $\langle P_i^t, P_i^v \rangle$ 通过基于mLLM的方法进行描述，其文本简洁且全面，即 $f2T: \langle P_i^t, P_i^v \rangle \rightarrow D_i$ 。同时，MAIC 采用知识提取方法来组织每页的核心知识并构建

用于幻灯片的树状分类法，即 $f3T: P_i^t, P_i^v, D_i \rightarrow K_j$ ，最终构成 P 。

3.1.2 计划阶段

在此阶段，教师（及授权助教）通过校对与润色成果参与其中。基于高度结构化的幻灯片，MAIC 构建了一种新型教学动作表征语言，可将讲座与提问等灵活教学功能融入预设课堂场景，该技术自然衔接相关教育技术如讲稿生成（Hong等人，2023b）与问题生成（Kurdi等人，2020）。同时，借助智能体构建技术（Chen 等人，2023），平台利用课程内容为教师与助教提供AI驱动智能体，助力后续课程基础架构的早期规划。

3. 功能生成。为使异构教学行为能够在课堂情境中生成，诸如授课和测验等教学活动被概念化为MAIC中的教学行为。每个教学行为 T 定义为 $T = (\text{类型}, \text{值})$ ，其中类型表示行为类别（例如ShowFile、ReadScript、AskQuestion），值则详细说明行为内容（如需朗读的脚本）。该方法体现了我们灵活性与适应性的原则，使课堂行为易于配置。它赋能开发者和教育工作者根据特定需求创建定制化教学行为，促进其与整体教学流程的无缝整合。

每个功能都与特定内容的生成相关联，记为 $\langle T_n, P_T \rangle$ 。例如，AskQuestion功能需要与一个或一组特定问题建立关联。其中最关键的步骤是ReadScript，因为它构成了教学过程的核心环节。基于这一功能，这些教学操作被嵌入到课程脚本中。

⁷以下文本中的类似符号也适用。⁸<https://openai.com/index/gpt-4v-system-card/>

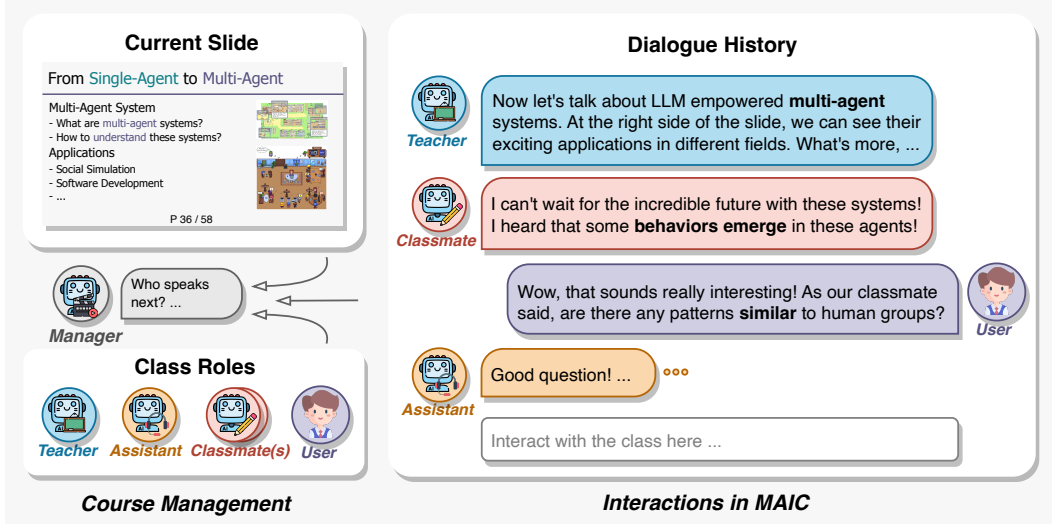


Figure 3: An illustration of the classroom learning environment of MAIC.

using special marker symbols, thereby enabling the intelligent agent to read and invoke them in a personalized manner. Specifically, MAIC has trained a high-quality lecture script generation model based on long-context encoding methods and multi-modal model foundations, thereby supporting the fundamental class procedures as well as the integration of other teaching actions, i.e., $f_T^4 : \hat{P} \rightarrow \hat{P}_{script}$. Then, MAIC provide a series of optional functions such as $f_T^5 : \hat{P} \rightarrow \hat{P}_{question}$ for serving proactive questioning during the class. Note that all the generated results are required to be checked and adjusted by instructors, which guarantees the quality and correctness of the produced course.

4. Agent Generation. Meanwhile, instructors can provide personalized information (such as the voice, teaching styles, and extended course material) for building customized teaching agents, such as Teacher Agent a_T and Teaching Assistant Agent a_{TA} . MAIC provides several agentization toolkits (Chen et al., 2023) that implemented via LLMs, supporting the high-level DIY of these agents. The extended course materials uploaded are also segmented in this section and integrated into different intelligent agents using the RAG (Retrieval-Augmented Generation) technology. The associated series of technological innovations are also thoroughly introduced and evaluated in the concurrent academic papers.

3.2 Learning: Multi-agent Classroom Environment

As described in the Introduction, student learning in MAIC follows a "*1 Student user + N AI Agents*" model. In such an environment, the AI teacher controls the learning progress based on the highly structured instructional action representation language mentioned earlier, explains course content, poses questions, and navigates PowerPoint slides, while the AI teaching assistant maintains classroom order and prevents content deviation. Students can interrupt the teacher at any time, ask questions, and engage in discussions, and the intelligent agents continuously adjust the teaching process and some content based on the students' performance. As introduce in Zhang et al. (2024), the design principles for constructing this immersive adaptive classroom originate from the following two concerns: (1) How to ensure that the classroom covers the core classroom behaviors? (2) How to maintain the entirety of the interaction within the natural flow of the classroom process?

In addressing the initial concern, we systematically classify classroom interaction behaviors in accordance with established educational principles, as delineated in Schwanke's seminal work (Schwanke, 1981): *Teaching and Initiation (TI)* encompasses the instructive actions of the teacher and the responsive feedback or insights provided by students; *In-depth Discussion (ID)* involves the alignment, deliberation, and iterative question-and-answer exchanges between the teacher and students, which are instrumental in facilitating students' conceptual comprehension; *Emotional Companionship (EC)* pertains to the encouragement of student learning, the cultivation of a conducive learning environment, and the provision of emotional sustenance; and *Classroom Management (CM)* refers to the



图3：MAIC 课堂学习环境示意图。

通过使用特殊标记符号，使智能代理能够以个性化方式读取并调用这些符号。具体而言，MAIC 基于长上下文编码方法和多模态模型基础，训练了一个高质量的讲稿生成模型，从而支持

基础课程程序以及其他教学活动的整合，即 $f4T: P \rightarrow$

$\wedge \quad \wedge \quad -P_{script}$ 。随后，MAIC 提供了一系列可选功能，例如 $f5T: P \rightarrow P_{question}$ ，用于在课堂中进行主动提问。需注意所有生成结果均需由教师检查和调整，以确保课程内容的质量与准确性。

4. Agent Generation .同时，教师可提供个性化信息（如语音、教学风格及扩展课程材料），用于构建定制化教学智能体，例如 Teacher Agent a_T 和 Teaching Assistant Agent a_{TA} 。MAIC 提供了多个基于LLMs实现的智能体化工具包（Chen 等人，2023），支持这些智能体的高级DIY开发。上传的扩展课程材料亦在此部分进行分段处理，并通过RAG（检索增强生成）技术整合至不同智能体中。相关系列技术创新也在同期学术论文中得到详尽阐述与评估。

3.2 学习：多智能体课堂环境

如引言所述，MAIC 中的学生学习遵循“1名学生用户 + NAI智能体”模型。在此环境中，AI教师基于前文提到的高度结构化教学动作表征语言控制学习进度、讲解课程内容、提出问题并浏览PowerPoint幻灯片，而AI教学助理则负责维持课堂秩序并防止内容偏离。学生可随时打断教师、提问并参与讨论，智能体会根据学生表现持续调整教学过程及部分内容。正如Zhang等人（2024）所述，构建这种沉浸式自适应课堂的设计原则源于以下两个考量：(1)如何确保课堂涵盖核心教学行为？(2)如何在课堂流程自然流动中保持互动完整性？

针对最初的关切，我们根据Schwanke开创性著作（Schwanke, 1981）中阐述的既定教育原则，系统地对课堂互动行为进行分类：**教学与引导（TI）**涵盖教师的指导性行为及学生提供的反馈或见解；**深度讨论（ID）**涉及师生之间的观点对齐、审议过程及迭代性问答交流，这对促进学生概念理解具有关键作用；**情感陪伴（EC）**涉及对学生学习的鼓励、良好学习环境的营造以及情感支持的提供；**课堂管理（CM）**指

maintenance of order, the organization of disruptive elements, and the steering of classroom discourse. Recognizing that these pedagogical behaviors manifest through diverse **Class Roles** (represented as $\widehat{\mathcal{R}} = \{r_i\}^{|\widehat{\mathcal{R}}|}$, with each r_i signifying a distinct role), it is imperative to ensure the *variety* and *breadth* of the agents' roles within the educational setting.

Addressing the subsequent concern, we emphasize the necessity of meticulously and rhythmically orchestrating the interactions among the various agents within the system, in harmony with the course content. With the Learning Materials (denoted as $C = [c_1, \dots, c_t]$, where each instructional script c_t is sequenced), we introduce an innovative **Session Controller** designed to regulate the flow of classroom interactions, contingent upon the class's dynamic state and under the aegis of a central managerial agent (Wu et al., 2023).

Based on these principles, we construct multiple classmate agents for diverse roles, implement class control, and ultimately derive the multi-agent classroom process.

Classmate Agents. To enhance the educational experience and emulate the dynamics of traditional classroom settings, we currently preset a variety of student-like agents, each imbued with unique personality traits, to complement the teaching agents. These agents are designed to perform roles akin to peer students, enriching the interactive landscape of the learning environment. In this scholarly work, we have introduced an initial set of four archetypal student agents, while also providing users with the flexibility to customize and introduce additional engaging student agents onto the educational platform. Each agent $\mathbf{a}_i \in \mathcal{A}$ is facilitated through prompting LLMs and associated with one or more class roles, denoted as: $\mathcal{A} = \rho(LLM, P_A)$, $\mathcal{A} \Leftrightarrow \widehat{\mathcal{R}}$, where ρ is the role customization operation, P_A is the system prompt with agent description (Zhang et al., 2024).

- *Class Clown (TI, EC, CM)*: Crafted to spark creativity, engender a lively classroom ambiance, and act as a supportive peer, this agent also assists the teacher in steering the class's focus when the learner's attention wanders.
- *Deep Thinker (TI, ID)*: This agent is dedicated to profound contemplation and to posing thought-provoking questions that challenge and extend the intellectual boundaries of the classroom.
- *Note Taker (TI, CM)*: With a penchant for summarizing and disseminating key points from the class discussions, this agent aids in the cognitive organization and retention of information for all participants.
- *Inquisitive Mind (TI, EC)*: Characterized by a propensity for inquiring about lecture content, this agent fosters a culture of inquiry and dialogue, prompting others to engage in critical thinking and collaborative discourse.

Unlike Standardized Operating Procedures (SOPs) commonly used in multi-agent systems (Qian et al., 2023; Hong et al., 2023a), classroom scenarios function as dynamic, interactive environments without rigid workflows, resembling an evolving group discussion. In these settings, agents must determine appropriate timing for their interactions, adapting to the fluid nature of classroom discourse. To address this need, we designed a controller that observes classroom dynamics, makes informed decisions, and manages agents' behaviors based on the current state of the class. The Session Controller is composed of three core modules: the Class State Receptor and Manager Agent.

Class State Receptor. The Class State Receptor captures the ongoing classroom dialogue, with the history up to time t represented as $H_t = \bigcup (u_i^{\mathbf{a}_j})^t$, where u_i is the utterance made by agent $\mathbf{a}_j$ or a user (denoted as $\mathbf{a}_u$). The class state S_t integrates this interaction data, structured as $S_t = \{P_t, H_t | \widehat{\mathcal{R}}\}$. Here, $P_t \subseteq P$ represents the learning materials covered up to time t . This design prioritizes adaptability and real-time decision-making, aligning with pedagogical principles that emphasize responsiveness to the evolving needs of learners within an educational setting.

Manager Agent. Drawing inspiration from AutoGen (Wu et al., 2023) and MathVC (Yue et al., 2024), we designed a hidden meta-agent responsible for regulating the dynamics of classroom interactions. This agent receives the current class state S_t , monitors the flow of the class, interprets ongoing activities, and determines the subsequent action to be executed, ensuring that the learning environment remains adaptive and responsive. The task $f_{\mathcal{L}}$ of the Manager Agent can be formally defined as $f_{\mathcal{L}} : \mathcal{S}_t \rightarrow (\mathbf{a}_t, \mathcal{T}) | \mathbf{a}_t \in \mathcal{A}, \mathcal{T}_n \Leftarrow \mathcal{T}$.

维持秩序、组织破坏性因素以及引导课堂讨论。认识到这些教学行为通过多样化的**班级角色**（以 $R = \{r_i\}_{i=1}^{\hat{R}}$ 表示，其中每个 r_i 代表一个独特角色）体现出来，因此必须确保教育环境中各主体角色的**多样性与广度**。

针对后续提出的关切，我们强调必须与课程内容保持协调，细致且有节奏地协调系统内各主体之间的互动。通过学习材料（标记为 $C = [c_1, \dots, c_t]$ ，其中每个教学脚本 c_t 按顺序排列），我们引入了一种创新的**会话控制器**，该控制器旨在根据课堂动态状态调节互动流程，并由中央管理主体负责统筹（Wu 等人，2023）。

基于这些原则，我们为不同角色构建了多类同学智能体，实现了班级管理功能，并最终推导出多智能体课堂运作流程。

同伴智能体。为提升教育体验并模拟传统课堂场景的互动动态，我们预先配置了多种具有独特人格特质的学生类智能体，以辅助教学智能体运作。这些智能体被设计为扮演类似同龄学生的角色，从而丰富学习环境中的互动场景。本研究中，我们首次引入了四类典型学生智能体，并为用户提供灵活定制功能，可将更多引人入胜的学生智能体引入教育平台。每个智能体 $a_i \in A$ 均通过提示式大语言模型实现，并关联一个或多个

角色角色，表示为： $A = \rho(LLM, P_A)$ ， $A \in \hat{R}$ ，其中 ρ 为角色定制操作， P_A 为包含智能体描述的系统提示（Zhang等人，2024）。

- **班级小丑 (TI, EC, CM)**：该角色旨在激发创造力、营造活跃课堂氛围并充当同伴支持者，当学生注意力分散时，还能协助教师引导班级注意力集中。
- **深度思考者 (TI, ID)**：该智能体专注于深度思考，并通过提出发人深省的问题来挑战并拓展课堂中的智力边界。
- **书记员 (TI, CM)**：该角色擅长总结并传播课堂讨论中的关键要点，有助于所有参与者对信息进行认知组织与记忆保留。
- **探究型思维 (TI, EC)**：该智能体以倾向于追问讲座内容为特征，能够培育探究与对话的文化氛围，促使他人参与批判性思维与协作性讨论。

与多智能体系统中常用的标准化操作规程（SOPs）不同（Qian等，2023；Hong等，2023a），课堂场景作为动态交互环境运作，没有固定的工作流程，类似于不断演化的小组讨论。在此类场景中，智能体必须根据课堂话语的流动性，确定互动的合适时机。为满足这一需求，我们设计了一种控制器，该控制器能够观察课堂动态、做出明智决策，并基于当前课堂状态管理智能体行为。会话控制器由三个核心模块组成：课堂状态接收器与管理智能体。

类状态接收器。类状态接收器捕捉课堂中的持续对话，时间 t 之前的对话历史表示为 $H_t = U(u_{ai})^t$ ，其中 u_{ai} 是智能体 a_j 或用户（记为 au ）的发言内容。类状态 S_t 整合这些交互数据，其结构定义为 $S_t = \{P_t, H_t | R\}$ 。此处 P_t 表示截至时间 t 所涵盖的学习材料。该设计优先考虑适应性和实时决策能力，符合强调对教育环境中学习者需求变化作出响应的教学原则。

Manager Agent。受AutoGen（Wu 等人，2023）和MathVC（Yue等人，2024）的启发，我们设计了一个负责调控课堂互动动态的隐藏元智能体。该智能体接收当前课堂状态 S_t ，监控课堂流程，解析正在进行的互动，并确定后续执行的动作，确保学习环境保持自适应性和响应性。管理智能体的任务 f_L 可形式化定义为 $f_L : S_t \rightarrow (a_t, T) | a_t \in A, T_n \in T$ 。

where $\mathcal{T}_n$ denotes a specific function, and the selected action will be carried out, transitioning the class to the next state. After executing an action, the system enters a waiting phase for a time window τ . During this period, if a user responds or the waiting time elapses, the Manager Agent is triggered to make a new decision. This design reflects key educational principles by prioritizing a learner-centered approach, maintaining fluid class engagement, and promoting timely and contextually relevant instructional adjustments, thereby enhancing the overall educational experience.

This classroom management method is the core of MAIC learning stage. Currently, we collect plenty interaction data and employ several foundation models (Hu et al., 2024; GLM et al., 2024) via fine-tuning or prompt tuning as our baseline model.

4 Key Technique Evaluation

MAIC is a complex large language model-based intelligent agent system that encompasses various specific technologies. On the teaching side, it involves multiple processes for content generation and knowledge understanding, while on the learning side, it requires evaluating the effectiveness of agent construction and classroom management capabilities. Currently, we focus on presenting two core functions: lecture script generation and course management, which are fundamental to the teaching and learning aspects of MAIC. The evaluation of other technologies will be continuously updated. It is important to note that these assessments provide only an initial view of specific aspects of MAIC rather than its overall effectiveness, which will be further explored through real-world practice and results analysis in subsequent sections.

4.1 Teaching Side Evaluation

Function Generation. As introduced in Section 3.1, generating vivid slide scripts is the core function of MAIC teaching workflow. **Baselines:** To assess the effectiveness of our implementation, we established two baseline configurations for comparison: (1) We replicated *Script2Transcript* (Nguyen et al., 2023), referred to as *S2T*, which uses slide titles to offer overarching context and guidance for generating transcripts. (2) We also reproduced *Self-Critique Prompting* (Zheng et al., 2024), denoted as *SCP*, which incorporates a self-critique and refinement process to enhance script quality.

Evaluation Metrics: Our evaluation employs four distinct metrics to rate the generated scripts on a 5-point Likert scale, where 1 represents unacceptable quality and 5 denotes optimal performance:

1. *Tone* evaluates whether the script appropriately reflects the instructional tone of a teacher.
2. *Clarity* measures how clear and comprehensible the script is for learners.
3. *Supportiveness* assesses the extent to which the script provides emotional and motivational support to students.
4. *Alignment* evaluates the degree to which the script content aligns with the slide material.

Overall performance is determined by averaging the scores across all metrics.

Procedure: We executed the course preparation pipeline for each baseline and collected script evaluations from annotators. To minimize potential bias, each slide was assessed by three independent annotators, who were required to provide ratings for all configurations of the same slide. This approach ensures a balanced and comprehensive evaluation of the teaching material, aligning with educational principles that emphasize clarity, support, and contextual relevance in instructional content.

Table 1: Evaluation result of our implementation of generation.

Setting	Tone	Clarity	Supportive	Matching	Overall
S2T (Nguyen et al., 2023)	3.88	3.93	3.23	3.63	3.67
SCP (Zheng et al., 2024)	4.03	<u>4.24</u>	3.38	3.93	<u>3.90</u>
MAIC-FuncGen	4.00	4.25	3.57	4.18	4.00
w/o visual	3.78	3.73	<u>3.44</u>	3.51	3.61
w/o context	3.97	4.00	3.38	<u>4.03</u>	3.84
Human	<u>4.02</u>	4.07	3.38	3.98	3.86

其中 T_n 表示特定函数，选定的操作将被执行，使班级进入下一状态。执行操作后，系统将进入等待阶段，持续时间窗口为 τ 。在此期间，若用户作出响应或等待时间到期，系统将触发管理代理进行新决策。该设计体现了核心教育原则：优先采用以学习者为中心的教学方法、保持课堂参与度的持续性，并促进及时且符合情境的教学调整，从而全面提升教育体验质量。

这种课堂管理方法是 MAIC 学习阶段的核心。目前，我们通过微调或提示调优收集了大量交互数据，并采用若干基础模型（Hu 等人，2024；GLM 等人，2024）作为基线模型。

4 关键技术评价

MAIC 是一个基于大型语言模型的复杂智能代理系统，整合了多种专业技术。在教学层面，该系统涉及内容生成与知识理解的多环节流程；在学习层面，则需要评估智能体构建效果及课堂管理能力。目前我们重点展示两大核心功能：讲义生成与课程管理，这些功能是 MAIC 教学与学习环节的基础支撑。其他技术指标的评估体系将持续更新完善。需要特别说明的是，当前评估仅能反映 MAIC 特定维度的初步表现，而非整体效能评估——后续章节将通过实际应用实践与结果分析，对系统整体效能展开深入探讨。

4.1 教学侧评

功能生成。如3.1节所述，生成生动的幻灯片脚本是 MAIC 教学 workflows 的核心功能。**基线：**为评估我们实现方案的有效性，我们建立了两种基线配置进行对比：(1)复现了 *Script2 Transcript*（Nguyen等人，2023），称为S2T，该方法利用幻灯片标题为转录生成提供整体上下文与指导；(2)同时复现了 *Self-Critique Prompting*（Zheng等人，2024），标记为SCP，该方法通过自我批评与优化流程提升脚本质量。

评估指标：本评估采用四项独立指标，对生成脚本进行五级李克特量表评分，其中1分代表质量不可接受，5分代表最佳性能表现：

1. *语调*用于评估脚本是否恰当体现了教师的教学语调。
2. *清晰度*用于衡量脚本对学习者的清晰度与可理解性。
3. *支持性*评估脚本对学生提供情感支持与动机激励的程度。
4. *对齐度*用于评估脚本内容与幻灯片素材的匹配程度。

整体性能通过所有指标评分的平均值来确定。

实验流程：我们为每个基准数据集执行了课程准备流程，并收集了标注人员的脚本评估结果。为最大限度减少潜在偏倚，每张幻灯片均由三位独立标注人员进行评估，且要求标注人员对同一幻灯片的所有配置方案均需给出评分。该评估方法确保了教学材料的平衡性与全面性，符合强调教学内容清晰性、支持性及情境相关性的教育原则。

表1：本研究中发电系统实施效果评估结果

设置	语调	清楚	支持性的	匹配	全面的
S2T（Nguyen等人，2023）	3.88	3.93	3.23	3.63	3.67
SCP（郑等人，2024）	4.03	<u>4.24</u>	3.38	3.93	<u>3.90</u>
MAIC-FuncGen	4.00	4.25	3.57	4.18	4.00
无视觉的	3.78	3.73	3.44	3.51	3.61
无上下文	3.97	4.00	<u>3.38</u>	<u>4.03</u>	3.84
人	<u>4.02</u>	4.07	3.38	3.98	3.86

Results: As shown in Table 1, our approach (MAIC-FuncGen) achieves the highest overall score of 4.00, outperforming all baseline methods. Our analysis reveals several key insights:

- **Importance of Visual Input:** The inclusion of visual elements significantly enhances script generation quality. Both *S2T* and MAIC-FuncGen without visual inputs received lower matching scores, highlighting the need for contextual visual cues that align the script content closely with the presented materials.
- **Role of Contextual Information:** The presence of coherent contextual information, including content from previous pages, is critical for script quality. It not only improves the clarity of the current content but also enhances the supportive and matching aspects by providing a continuous and interconnected learning narrative. This aligns with pedagogical principles that emphasize coherence and context in learning materials to support deeper understanding.
- **Comparative Performance with Human Instructors:** Interestingly, our approach slightly outperformed the human baseline across three key dimensions. This can be attributed to the inherent ability of large language models (LLMs) to strictly adhere to instructions, maintaining alignment with slide content and employing an encouraging and supportive tone. In contrast, human instructors often expand on topics freely, introducing their own style and diverging from the core content, which reflects a more flexible, albeit less structured, instructional approach.

These findings underscore the value of integrating structured visual and contextual information into script generation, while also highlighting the potential of LLMs to complement traditional instructional strategies by providing consistency and structured support.

4.2 Learning Side Evaluation

Classroom Manager Agent. Section 3.1.2 also mention several relevant techniques of MAIC learning. The classroom manager agent is the keypoint of the classroom controlling. **Setup:** However, the evaluation of this process is highly subjective, making it challenging to establish an objective scoring system for assessment. Therefore, in the practical implementation of the TAGI and HSU courses, we select 500 actual system decisions and extracted their corresponding classroom scenarios. We recruit expert teachers and teaching assistants to manually annotate these scenarios. Based on this annotated data, we derive the results shown in Figure 4. These results illustrate the alignment between the actions chosen by the manager agent and those selected by human instructors in determining the next course action. Specifically, we evaluate the implementation with and without role description, detecting the effects of these contextual information.

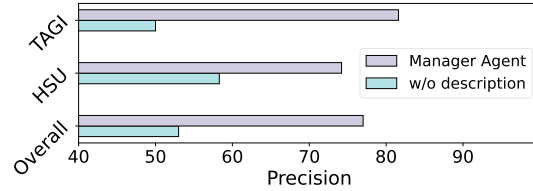


Figure 4: Manager Agent Precision.

Result: Statistical analysis reveals that omitting role descriptions for each agent reduces the classifier’s performance. Although the LLM can sometimes identify the correct agent by referencing partial behaviors from the chat history, the inclusion of comprehensive role descriptions markedly enhances performance. This suggests that while leveraging chat history as input for the scene controller can provide some benefits, it is insufficient for consistently generating accurate outputs.

The current results, however, remain below optimal levels, indicating further opportunities to refine and enhance the user experience. Despite the suboptimal performance, interacting agents demonstrate the capacity to partially offset these shortcomings. This compensatory effect is due to the LLM’s ability to manage user queries beyond the predefined functions of each agent, as evidenced by our subsequent behavioral study, where user ratings did not significantly decline in the ablation setting.

Nonetheless, enhancing the accuracy of the controller agent remains advantageous, as agents can more effectively manage tasks they are specifically designed for. For example, the teacher agent is tailored to adopt a softer, more instructive tone, but it may be less effective in handling safety-related cases compared to the teaching assistant agent. Improved accuracy ensures that each agent operates within its designed scope, contributing to a more seamless and effective instructional process.

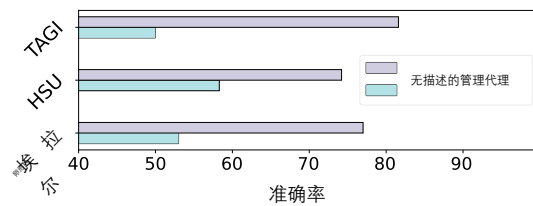
结果：如表1所示，我们的方法（MAIC -FuncGen）取得了最高的总体分数4.00，优于所有基线方法。分析结果揭示了若干关键发现：

- **视觉输入的重要性：**视觉元素的加入能显著提升脚本生成质量。无论是S2T还是MAIC -FuncGen模型，若缺乏视觉输入都会获得较低的匹配分数，这凸显了需要通过上下文视觉线索使脚本内容与呈现材料高度匹配的重要性。
- **上下文信息的作用：**连贯的上下文信息（包括前文内容）对脚本质量至关重要。它不仅能提升当前内容的清晰度，还能通过构建连续且相互关联的学习叙事，增强内容的支撑性和衔接性。这与教学原则高度契合——教学材料需要强调连贯性和上下文关联性，才能有效促进学习者的深度理解。
- **与人类导师的对比表现：**有趣的是，我们的方法在三个关键维度上略微优于人类基准模型。这主要归因于大型语言模型（LLMs）固有的能力——能够严格遵循指令要求、保持与幻灯片内容的关联性，并采用鼓励性与支持性的语气。相比之下，人类导师往往会在讲解过程中自由拓展话题，融入个人风格并偏离核心内容，这种教学方式虽然更具灵活性，但结构化程度相对较低。

这些研究结果凸显了在脚本生成过程中整合结构化视觉信息与上下文信息的价值，同时也表明大型语言模型（LLMs）通过提供一致性与结构化支持，能够有效补充传统教学策略的潜力。

4.2 学习侧边评价

课堂管理智能体。第3.1.2节还提及了若干MAIC学习的相关技术。课堂管理智能体是课堂调控的核心要素。**系统配置：**然而该过程的评估具有高度主观性，导致难以建立客观的评分体系。因此在TAGI与HSU课程的实际实施中，我们选取了500个真实系统决策案例，并提取其对应的课堂场景。我们邀请专家教师参与评估——图4：管理智能体精度指标。



我们安排研究人员和教学助理对这些场景进行人工标注。基于标注后的数据，我们得出了图4所示的结果。这些结果清晰展现了管理代理选择的动作方案与人类教师在确定下一步教学行动时所选方案之间的高度一致性。具体而言，我们通过对比包含角色描述与不包含角色描述两种实施方案，系统评估了这些情境信息对教学决策的影响效果。

结果：统计分析表明，若省略每个智能体的角色描述会降低分类器的性能表现。尽管大语言模型（LLM）有时能通过参考聊天记录中的部分行为来识别正确智能体，但完整角色描述的加入能显著提升分类效果。这说明虽然利用聊天记录作为场景控制器的输入源能带来一定优势，但仅凭此仍难以持续生成准确的输出结果。

然而当前实验结果仍低于理想水平，表明在优化用户体验方面仍有提升空间。尽管表现未达最优状态，交互式智能体仍展现出部分弥补这些缺陷的能力。这种补偿效应源于大语言模型（LLM）能够处理超出各智能体预设功能范围的用户查询——后续行为学研究证实，在消融实验设置中用户评分并未出现显著下降。

尽管如此，提升控制器智能体的准确性仍具有显著优势，因为智能体能更高效地完成其专为特定场景设计的任务。例如，教师智能体被设计为采用更温和、更具指导性的沟通方式，但在处理安全相关案例时，其效果可能不及教学助理智能体。精准度的提升确保每个智能体都能在其设计范围内运作，从而为教学过程提供更无缝衔接且高效能的支持。

5 Behavioral Experiment

Following approval from Tsinghua University Science and Technology Ethics Committee (Certificate No: THU-04-2024-56) and the recruitment of student and teacher volunteers, we conduct over three months of teaching practice and behavioral analysis in the courses "Towards General Artificial Intelligence" and "How to Study in the University." This large-scale study involves more than 500 students and aims to address three core questions: **Q1:** What is the quality of MAIC Courses? **Q2:** What are the learning outcomes within MAIC? **Q3:** How do students perform in the MAIC environment? In the following sections, we present some preliminary observations from this study.

5.1 Q1: The Quality of MAIC Course

We evaluated the quality of the MAIC course using the results from two questionnaires completed by course takers. The first questionnaire focused on the quality of AI teacher’s teaching, adapted from the Community of Inquiry Framework (Garrison and Arbaugh, 2007). Items in the original COI questionnaire were revised to make them suitable for the AI-engaged learning environment. This questionnaire was administered when students completed the whole course.

As presented in Figure 5, the results showed that overall students had positive beliefs of the teaching quality on MAIC. For instance, the mean score of students’ rating on the question "The AI instructor clearly communicated important course goals" (Course Objective) was 4.12 (SD = 0.66) out of 5, and the average rating on "The AI instructor encouraged course participants to explore new concepts" was 4.03 (SD = 0.73). These findings suggest that students felt the AI instructors effectively helped them understand course objectives, clarify their thinking, explore new ideas, and engage in meaningful dialogue.

However, relatively lower ratings were observed on questions like "The AI instructor provided feedback that helped me understand my strengths and weaknesses" (Understand Student), which had a mean score of 3.51 (SD = 0.94). This indicates that AI instructors may lack personalization and adaptability during the teaching process, possibly because the same scripts were used for all students.

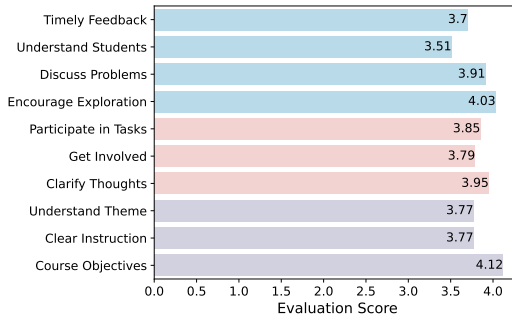


Figure 5: Results from after-course survey.

5.2 Q2: The Behavior of MAIC Student Engagement

Firstly, when it comes to choosing the class mode, students tend to prefer the "continuous mode," believing that this mode allows them to maintain their train of thought without interruption, thereby ensuring learning efficiency. The "continuous mode" refers to a setting where, after selecting the teacher and other intelligent agent roles (such as teaching assistant, thinker, note-taker), the chosen roles conduct the class from start to finish without any interactive input from the students, making it a relatively passive learning approach. For example, during interviews, one student mentioned:

"I mostly used the continuous mode because, in the interactive mode, after the AI teacher finishes each sentence, you have to respond before they can continue. I don't always feel like interacting after every sentence, so most of the time, I use continuous mode. Of course, there were one or two times when I used interactive mode because the AI prompted me to speak, but I remember that once, after I spoke in interactive mode, the AI didn't respond or react to what I said, so I felt like it wasn't very useful. After that, I just stuck with the continuous mode." In this mode, although interaction is not possible, some students adopt a pause strategy if they don't understand something. For instance, one student mentioned, "If I didn't understand something, I would pause and look at the PPT and the text in the text box. I don't think I ever stopped to ask the AI to explain something again, unless it was some unfamiliar term or a more exploratory topic."

5 行为实验

在获得清华大学科学技术伦理委员会批准（证书编号：THU-04-2024-56）并招募学生与教师志愿者后，我们在《通识人工智能导论》和《大学学习之道》课程中开展了为期三个月的教学实践与行为分析。这项大规模研究涉及500余名学生，旨在解答三个核心问题：**问题1**：MAIC 课程质量如何？**问题2**：MAIC 内学习成果有哪些？**问题3**：学生在 MAIC 环境中的表现如何？下文将呈现本研究的部分初步观察结果。

5.1 问题1：MAIC 课程的质量

我们通过学员填写的两份问卷评估了 MAIC 课程的质量。第一份问卷聚焦于人工智能教师的教学质量，改编自探究性学习社区框架（Garrison 与 Arbaugh, 2007）。原COI问卷中的条目经过修订，以适应人工智能参与式学习环境。该问卷在学生完成全部课程后发放。

如图5所示，研究结果显示学生对 MAIC 教学质量普遍持积极态度。例如，在“AI讲师清晰传达了重要课程目标”（课程目标）这一问题中，学生平均评分为4.12分（标准差=0.66，满分5分）；而在“AI讲师鼓励学员探索新概念”这一问题上，平均评分为4.03分（标准差=0.73）。这些数据表明，学生认为AI讲师有效帮助他们理解课程目标、理清思维脉络、开拓创新思路，并促进了富有成效的学术交流。

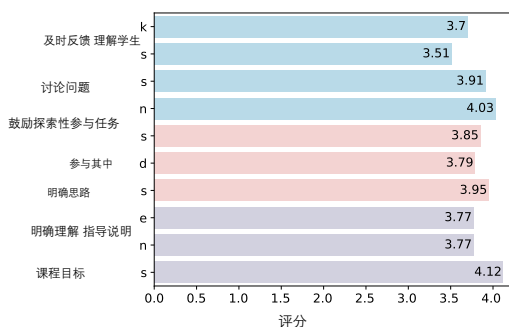


图5：课程后调查结果。

然而，在诸如“AI导师提供的反馈帮助我理解了”等问题上观察到相对较低的评分。

“优势与劣势”（学生理解度）的平均得分为3.51（标准差=0.94）。这表明人工智能教学系统在教学过程中可能缺乏个性化与适应性，原因可能是所有学生均使用了相同的教学脚本。

5.2 Q2: MAIC 学生参与度的行为特征

首先，在选择课堂模式时，学生们普遍倾向于选择“连续模式”，认为这种模式能让他们保持思维连贯性而不受干扰，从而确保学习效率。所谓“连续模式”是指在选定教师及其他智能角色（如助教、思考者、笔记员）后，这些角色将全程主导课堂活动，学生无需进行任何互动输入，属于相对被动的学习方式。例如在访谈中，有学生提到：

我主要使用连续模式，因为在交互模式下，AI教师说完每句话后，你必须先回应才能继续。我并不总想在每句话后都进行互动，所以大多数时候我选择连续模式。当然，也有过一两次使用交互模式的情况，因为AI会提示我发言，但我记得有一次在交互模式发言后，AI没有回应或对我所说的内容做出反应，因此觉得效果不佳。之后我就坚持使用连续模式了。“在这种模式下，虽然无法进行互动，但有些学生若理解有误会采用暂停策略。例如有学生提到：“如果我不明白某些内容，我会暂停并查看PPT和文本框中的文字。除非遇到陌生术语或更具探索性的话题，否则我从未停下来要求AI再次解释。”

Secondly, regarding specific behaviors during the class, some students proactively ask questions to the intelligent agent around certain topics. As shown in Figure 3, 61% of the students' behavior in class involved actively seeking knowledge, information, or asking questions. For example, asking "Can you explain the transformer structure in simple terms?" The interview results of this study also support this view. Many students indicated that they would actively ask questions. Some examples from the interviews include:

- "When I asked a question, the AI would ask a follow-up question related to mine, and I felt it was an amazing experience, like it was really guiding me to think more deeply. I think the questions it asked made a lot of sense. In that interactive mode, it could extend into many other discussions beyond the course content."
- "I think this might be an advantage of AI teaching. Because, in a traditional classroom, whether it's a large class or even a small one, students nowadays are generally reluctant to ask questions. There are various reasons for this. Also, I think immediate Q&A helps me learn the material better. If I have a question, I can get an answer right away, and I think that immediate feedback is really valuable."

Additionally, some students manage and control different intelligent agent roles and the class progress. These management-related behaviors account for 11% (Figure 6). For example, "Please go back to the previous slide," or "Please explain that in simpler terms," demonstrating strong autonomy and self-regulated learning abilities. In the interviews, some students also mentioned, *I don't just ask for knowledge; I might ask, 'I want to learn more,' or 'I hope to explore something new in a certain field,' to manage and regulate the AI's responses.*

Overall, the findings reveal that while students prefer the "continuous mode" for its uninterrupted flow of information, this passive approach may limit opportunities for active engagement and critical thinking. In the future, AI agent-driven classes should actively encourage student interaction rather than simply delivering continuous knowledge. In addition, the high level of proactive questioning indicates that students are eager to engage when given the opportunity, underscoring the importance of designing AI tools that foster inquiry-based learning (Xie, 2023). Questioning is an important behavior that reflects active learning in students, which ultimately leads to better academic performance. Finally, the occurrence of management-related behaviors suggests that students already realize their active role in their learning process, which align with Cain (2024)' study. It is said that the capabilities of LLM AI tools can help students and educators transition from passive recipients to active co-creators of their learning experiences. Further research can give more support and encouragement for them to take an active role in directing their learning.

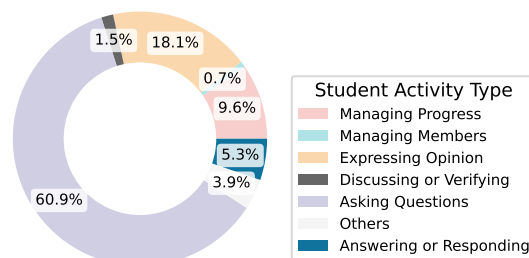


Figure 6: Ratio of student activities.

5.3 Q3: The Outcome of MAIC Learning

We assessed the effectiveness of the MAIC course from three perspectives: performance in module tests and the final exam, technology acceptance through questionnaires, and self-reported higher-order thinking scores.

Test Results. Module tests were conducted at the end of each module, focusing primarily on the content covered in the most recent module. The final exam was administered one week after the course concluded, synthesizing questions from the earlier module tests. Average attendance of the module tests is 76.3% (SD=6%), while that of the final exam is 73.3%. Test scores (standardized to percentage) ranged from 53.3% (Module 2, SD=18.9%) to 82.4% (Module 4, SD=16.3%), reflecting students' learning outcomes. These outcomes were corroborated by student interviews. One participant expressed, "My most impressive gains, from the perspective of knowledge, all came from the post-class test. If there were no post-class test, I might not remember the knowledge at all, and it might just pass by like a passing cloud of smoke... When I took the test and then looked back at the courseware, it was the peak period for me to absorb knowledge, honestly speaking."

其次，关于课堂上的具体行为表现，部分学生会主动向智能助手就特定主题提问。如图3所示，61%的学生课堂行为涉及主动寻求知识、信息或提问。例如会提出“能否用简单语言解释变压器结构？”这类问题。本研究的访谈结果也印证了这一现象，许多学生表示会主动提问。访谈中的一些典型案例包括：

- 当我提出问题时，AI系统会随即生成与之相关的追问，这种互动体验令人惊叹，仿佛它在引导我进行更深层次的思考。系统提出的问题极具逻辑性，在这种互动模式下，讨论内容甚至能延伸至课程大纲之外的诸多领域。
- 我认为这可能是人工智能教学的优势所在。因为在传统课堂中，无论是大班还是小班，如今的学生普遍不愿提问。造成这种现象的原因多种多样。此外，我认为即时问答有助于我更好地掌握教学内容。当我遇到疑问时，能够立即获得解答，这种即时反馈确实具有重要价值。

此外，部分学生能够自主管理并调控不同智能体角色及课堂进程。这类管理行为占比达11%（图6）。例如“请返回上一张幻灯片”或“请用更简单的语言解释”等指令，充分展现了其高度自主性和自我调节学习能力。访谈中，有学生还提到：我不仅主动寻求知识，还会提出‘我想进一步学习’或‘希望探索特定领域的新方向’等诉求，以此来调控人工智能系统的响应内容。

总体而言，研究结果表明，尽管学生更倾向于选择“连续模式”以获取不间断的信息流，但这种被动学习方式可能限制主动参与和批判性思维的机会。未来由AI智能体主导的课堂应积极鼓励学生互动，而非单纯传递连续性知识。此外，学生表现出的高度主动提问行为表明，只要给予机会他们就渴望参与学习，这凸显了设计促进探究式学习的AI工具的重要性（Xie, 2023）。提问是一种关键学习行为，这体现了学生主动学习的积极态度，最终能提升学业表现。值得注意的是，管理相关行为的出现表明，学生们已充分意识到自身在学习过程中的主体性角色，这一发现与凯恩（2024）的研究结论高度吻合。研究表明，LLM人工智能工具的能力能够帮助师生从被动接受者转变为学习体验的主动共建者。未来研究可进一步为学生提供支持与激励，助力他们主动主导学习进程。

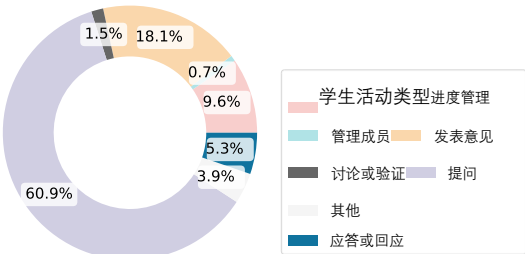


图6：学生活动比例

5.3 问题3：MAIC 学习的效果

我们从三个维度评估了 MAIC 课程的效果：模块测试与期末考试的表现、问卷调查反映的技术接受度，以及自我报告的高阶思维能力得分。

测试结果。各模块测试均在课程结束时进行，重点考查最新模块所涵盖的内容。期末考试于课程结束后一周实施，综合了前期模块测试中的题目。模块测试平均出勤率为76.3%（标准差=6%），期末考试出勤率为73.3%。标准化后的测试分数范围为53.3%（模块2，标准差=18.9%）至82.4%（模块4，标准差=16.3%），充分体现了学生的学习成效。学生访谈结果进一步佐证了这些学习成果。一位参与者表示：“从知识掌握程度来看，我最显著的进步都来自课后测试。如果没有课后测试，我可能完全记不住知识点，就像过眼云烟般转瞬即逝……说实话，当我参加测试并回顾课程资料时，正是我吸收知识的黄金时段。”

Additionally, test scores are strongly associated with class engagement. Specifically, the frequency (measured by the logarithm of the number of messages per module) and length of in-class chat messages (measured by the logarithm of the number of characters per message and module)—a prominent feature of the MAIC system—were positively correlated with standardized test scores and final exam performance, as presented in Table 2. A regression analysis of standardized test scores on in-class chat message metrics was performed, controlling for normalized scores from Module 1. As Table 2 shows, in-class chat engagement is found to significantly predict higher test scores.

Table 2: Correlation of students’ message-aware behavior and test results. Values shown in the table are normalized.

w/o control	$\mu(\log(\text{MsgNum}))$	$\mu(\log(\text{MsgLen}))$
AvgQuiz	0.341***	0.202*
FinalExam	0.346***	0.333**
w/ control	$\mu(\log(\text{MsgNum}))$	$\mu(\log(\text{MsgLen}))$
AvgQuiz	0.206**	0.177**
FinalExam	0.174	0.235*

Technology Acceptance. We further evaluate the acceptance of generative AI tools, such as ChatGPT, before and after the course. The results demonstrated a significant overall increase in technology acceptance ($N = 111$, $t = 3.05$, $p = 0.002$). Further analysis of specific dimensions of acceptance revealed significant improvements in Habit ($t = 2.81$, $p = 0.005$), Effort Expectancy ($t = 3.98$, $p < 0.001$), and Facilitating Conditions ($t = 3.22$, $p = 0.002$). These findings indicate that students grew increasingly accustomed to and supportive of the MAIC course format. Interview responses also reflected an enhanced understanding and acceptance of AI technologies. One student remarked:

"I used to be quite resistant to AI, mainly because it was too complicated, but after this course, I found that it was not that complicated, and I became more accepting of it. I also learned some of its principles."

High-order Thinking. We explore the perceived impact of using LLMs on students’ higher-order thinking skills through pre- and post-course questionnaires. Comparative analyses and t-test results showed significant increases in students’ perceptions of the positive effects of the course on their abstract thinking ($t = 2.32$, $p = 0.02$) and critical thinking ($t = 2.37$, $p = 0.02$). These findings suggest that students believe using large language models throughout the course enhanced their cognitive abilities in these areas. Interviews further illuminated these perceptions. One student stated:

"I think it might make me more confident in asking these questions and think more," while another mentioned, "Unlike before, (I was always) hiding and waiting for the opportunity to ask again."

However, the course’s impact on other higher-order thinking skills remained ambiguous. Several students noted a lack of deep thinking and discussion opportunities in the course. For instance, one student commented:

"(In a real classroom) After class, I can ask the teacher to explain it to me again... At this time, the teacher will definitely give you different ideas or explanations, but in this class, you will have no more after class." Another student added, "this course is purely about content because the teacher’s teaching is quite mechanical. The course is rich in theoretical knowledge, but there is almost no life thinking, life enlightenment, or some enlightenment and thinking outside of artificial intelligence."

Notably, there is a limitation came from the questionnaire itself. As we evaluated students’ perceptions on the impacts on high-order thinking abilities, it should be noted that the abilities are not estimated. Future study should include designed scales or tasks to estimate the effects of the course on the students’ high-order thinking abilities.

To address this limitation, future research should incorporate well-designed scales or specific tasks that can objectively measure students’ higher-order thinking abilities. These might include assessments that evaluate critical thinking, problem-solving, and analytical reasoning skills directly. By doing so, researchers can gain a more comprehensive understanding of the course’s effectiveness in enhancing

此外，考试成绩与课堂参与度呈显著相关性。具体而言，课堂聊天信息的发送频率（以每模块消息数量的对数为衡量指标）及聊天信息长度（以每条消息和每个模块的字符数量对数为衡量指标）均具有统计学意义。

MAIC 系统的一个显著特征——与标准化测试分数及fi呈正相关

如表2所示，未观察到考试成绩变化。我们对标准化测试分数与课堂聊天信息指标进行了回归分析，并控制了模块1的标准化分数。表2显示，课堂聊天参与度可显著预测更高的测试分数。

技术接受度。我们进一步评估了课程前后对生成式AI工具（如ChatGPT）的接受度。结果显示技术接受度整体显著提升（ $N=111$ ， $t=3.05$ ， $p=0.002$ ）。对接受度具体维度的进一步分析表明，习惯性使用（ $t=2.81$ ， $p=0.005$ ）、努力预期（ $t=3.98$ ， $p<0.001$ ）以及促进条件（ $t=3.22$ ， $p=0.002$ ）均呈现显著改善。这些发现表明学生对 MAIC 课程形式的适应性与支持度持续增强。访谈反馈也反映出对AI技术的理解与接受度提升。一名学生表示：

我过去对人工智能技术持强烈抵触态度，主要因其结构过于复杂。但完成本课程后，发现其实际应用并不复杂，因而对其接受度显著提升。同时，我也系统掌握了相关技术原理。

高阶思维。我们通过课程前后问卷调查，探究使用大语言模型对学生高阶思维能力的影响感知。比较分析与t检验结果表明，学生对课程在抽象思维（ $t=2.32$ ， $p=0.02$ ）和批判性思维（ $t=2.37$ ， $p=0.02$ ）方面产生的积极影响感知显著提升。这些发现表明，学生认为在整个课程中使用大语言模型增强了他们在这些领域的认知能力。访谈进一步阐明了这些认知。一名学生表示：

我认为这可能会增强我提出这些问题的信心，并促使我进行更多思考，”而另一位受访者则表示，“与以往不同，（我过去总是）隐藏着并等待再次提问的机会。”

然而，该课程对其他高阶思维能力的影响仍不明确。多名学生指出课程中缺乏深度思考与讨论机会。例如，一名学生评论道：

（在真实的课堂环境中）下课后我可以请老师再给我讲解一遍.....此时老师肯定会给出不同的见解或解释，但在这门课中，下课后你将再也无法获得这些内容。
“另一位学生补充道：”这门课程纯粹是知识灌输，因为老师的教学方式相当机械。课程内容虽然理论知识丰富，但几乎缺乏生活思考、人生启迪，也缺少人工智能领域之外的启发性思考。”

值得注意的是，问卷本身存在局限性。由于我们评估的是学生对高阶思维能力影响的认知，需特别说明的是，这些能力并未被量化评估。未来研究应采用设计化的量表或任务，以评估课程对学生高阶思维能力的影响。

为解决这一局限性，未来研究应采用设计精良的量表或特定任务，以客观评估学生高阶思维能力。此类评估可直接考察批判性思维、问题解决及分析推理能力。通过这种方式，研究者能更全面地了解该课程在提升学生能力方面的有效性。

表2：学生信息意识行为与测试结果的相关性。表中数值已进行标准化处理。

无控制	$\mu(\log(\text{MsgNum}))$	$\mu(\log(\text{MsgLen}))$
测验	0.341***	0.202*
期末考试	0.346***	0.333**
带控制的	$\mu(\log(\text{MsgNum}))$	$\mu(\log(\text{MsgLen}))$
测验	0.206**	0.177**
期末考试	0.174	0.235*

these abilities. Moreover, the inclusion of such measures would allow for a more rigorous evaluation of the course's impact, providing stronger evidence to support or challenge the findings based on student perceptions alone.

6 Predicted Impact and Ethical Consideration

6.1 Predicted Impact

The implementation of MAIC in online education is expected to revolutionize the learning experience by enhancing both scalability and adaptability. By leveraging multi-agent systems, MAIC can dynamically adjust to the needs of individual learners, providing personalized learning paths that were previously unattainable in traditional MOOCs. This personalized approach not only improves learning outcomes but also provides access to high-quality education across diverse socio-economic backgrounds, as it reduces the dependency on human instructors for content delivery.

Moreover, MAIC is anticipated to address some of the inherent challenges of traditional online education, such as the one-size-fits-all model and the lack of real-time adaptability (Rizvi et al., 2022). The integration of AI-driven agents as teachers, teaching assistants, and classmates creates a more interactive and responsive learning environment. This shift promises to increase engagement and motivation among students, potentially leading to higher completion rates and deeper understanding of the material.

However, it is crucial to acknowledge that the introduction of such a transformative system could also have unintended consequences. There may be a widening gap between students who adapt well to AI-powered learning environments and those who struggle with this new mode of education. Additionally, the reliance on AI systems could lead to reduced opportunities for human instructors, potentially diminishing the role of educators in the learning process. These impacts need to be carefully monitored and addressed through ongoing evaluation and refinement of the MAIC system.

6.2 Ethical Considerations

The deployment of MAIC in online education brings forth significant ethical considerations that must be carefully evaluated (Almarzouqi et al., 2024). One of the foremost concerns is learner privacy and data security. The system's reliance on large-scale data collection and analysis to personalize learning experiences raises questions about how student data is stored, accessed, and used. To mitigate these concerns, stringent data protection measures have been implemented, including encryption and anonymization of student records. However, given the sensitivity of educational data, continuous efforts to enhance security protocols are essential.

Issues of discrimination and bias also present ethical challenges. While MAIC aims to provide a personalized learning experience for all students, there is a risk that the algorithms driving these personalized experiences may inadvertently reinforce existing biases, particularly if the training data is not sufficiently diverse. To address this, the development team has incorporated fairness-focused auditing procedures into the algorithmic design process. Although these measures are designed to minimize bias, it is recognized that no system is entirely immune to these challenges, and ongoing monitoring is required to ensure equitable outcomes.

The accuracy of information and content regulation within MAIC is another critical area of ethical concern. As the system automates the creation and dissemination of educational content, there is a risk that inaccuracies could be propagated at scale. To mitigate this, content generated by the AI is regularly reviewed by subject matter experts and teaching assistants. Nonetheless, given the vast scale of content production, it is acknowledged that some errors may still occur. Thus, the system includes mechanisms for students and educators to flag and correct inaccuracies, ensuring continuous improvement of the educational material.

In terms of the ethics of education, MAIC raises questions about the role of teachers in a system that increasingly relies on AI-driven instruction. While the system enhances scalability and provides personalized learning experiences, it also diminishes the direct involvement of human educators. This shift may impact the development of teacher-student relationships, which are critical for fostering emotional and social growth in learners (Thornberg et al., 2022). To address this, MAIC includes human-in-the-loop design principles, ensuring that educators can intervene and guide the AI's

这些能力。此外，纳入此类评估指标将有助于更严格地评估课程效果，从而为仅基于学生主观感受得出的研究结论提供更有力的支持或质疑依据。

6 预测影响与伦理考量

6.1 预测影响

在线教育中 MAIC 的实施有望通过提升可扩展性和适应性来彻底改变学习体验。借助多智能体系统，MAIC 能够动态响应个体学习者的需求，提供传统慕课难以实现的个性化学习路径。这种个性化教学模式不仅能提升学习效果，还能让不同社会经济背景的学习者获得优质教育资源，因其减少了对人工讲师内容传授的依赖。

此外，MAIC 有望解决传统在线教育中的一些固有挑战，例如一刀切的教学模式和缺乏实时适应性（Rizvi 等人，2022）。通过将人工智能驱动的智能体整合为教师、助教和同学，可构建更具互动性和响应性的学习环境。这种转变有望提升学生参与度与学习动机，从而可能提高课程完成率并深化对教学内容的理解。

然而必须清醒认识到，这种变革性系统的引入也可能带来意想不到的后果。那些能适应人工智能驱动学习环境的学生与难以适应这种新型教育模式的学生之间，可能会出现日益扩大的差距。此外，过度依赖人工智能系统可能导致人类教师的职业机会减少，进而削弱教育工作者在学习过程中的作用。这些潜在影响需要通过持续评估和优化 MAIC 系统来密切关注并妥善应对。

6.2 伦理考量

MAIC 在在线教育中的应用引发了诸多重大伦理考量，必须进行审慎评估（Almarzouqi 等，2024）。首要关切在于学习者隐私与数据安全问题。该系统依赖大规模数据采集与分析来实现个性化学习体验，这引发了关于学生数据存储、访问及使用方式的质疑。为缓解这些担忧，已实施包括学生档案加密与匿名化在内的严格数据保护措施。然而鉴于教育数据的敏感性，持续完善安全协议仍至关重要。

歧视与偏见问题同样带来伦理挑战。虽然 MAIC 致力于为所有学生提供个性化学习体验，但驱动这些个性化体验的算法存在无意强化既有偏见的风险，特别是在训练数据缺乏足够多样性的情况下。为解决这一问题，开发团队已在算法设计过程中引入以公平性为核心的审计机制。尽管这些措施旨在最大限度减少偏见，但业界普遍认识到任何系统都无法完全规避此类挑战，仍需持续监控以确保结果公平性。

MAIC 系统中信息准确性与内容规范管理是另一个关键的伦理关注领域。随着系统自动化生成并传播教育内容，存在不准确信息大规模传播的风险。为降低这种风险，人工智能生成的内容会定期接受学科专家和教学助理的审核。然而，考虑到内容生产规模庞大，仍需承认某些错误仍可能发生。为此，系统特别设置了学生与教师标记并修正错误的机制，确保教育材料的持续优化。

在教育伦理学层面，MAIC 引发了关于教师角色的思考——在一个日益依赖人工智能驱动教学的体系中，教师究竟扮演着怎样的角色？虽然该系统提升了可扩展性并提供了个性化学习体验，却也削弱了人类教育者的直接参与度。这种转变可能影响师生关系的发展，而师生关系对促进学习者的情感与社会成长至关重要（Thornberg 等，2022）。为解决这一问题，MAIC 引入了“人机协同设计”原则，确保教育者能够及时介入并指导人工智能系统。

decision-making processes when necessary. However, the balance between AI automation and human oversight remains a complex issue that requires ongoing consideration.

Furthermore, the lack of peer interaction in a predominantly AI-driven educational environment could hinder the development of important social skills among students. To counter this, MAIC incorporates AI classmates designed to simulate peer interactions. While these agents provide a form of interaction, they cannot fully replicate the nuances of human-to-human communication. The system, therefore, encourages mixed-mode learning environments where students can engage with both AI and human peers, preserving the benefits of social learning.

Finally, the issue of student personalization must be approached with caution. While MAIC’s ability to adapt to individual learning needs is a significant advantage, it also poses risks to fairness and equality. There is a potential for certain students to receive more tailored and effective instruction based on their data profiles, potentially exacerbating educational inequalities. To mitigate this, MAIC includes mechanisms to ensure that all students, regardless of their data profiles, have access to high-quality learning experiences. The system’s fairness algorithms are continuously refined to promote equitable educational outcomes for all learners.

7 Conclusion

In this paper, we provide a concise overview of the development trajectory of online education and the technological opportunities arising in the era of large language models. Considering the principles of adaptivity and scalability, along with the sophisticated design of LLM-driven multi-agent systems, we explore how existing MOOC can be transformed into MAIC (Massive AI-empowered Courses) and discuss new paradigms of teaching and learning. We propose a comprehensive solution, analyze the key technical components, and implement each step of the process. Our approach was practically deployed in two courses at Tsinghua University, leading to a series of preliminary observations of student behavior. These initial findings suggest that highly personalized classrooms built with new AI-assisted learning technologies can achieve high quality, and student behavior demonstrates the effectiveness of the teaching process. Moving forward, this work will be continuously maintained and expanded, aiming to develop an open and shared platform for educational exploration, academic research, and technological innovation. We hope our work will call upon and serve educational theorists, technology developers, and innovators to engage in discussions about the new online environment in the era of large language models.

Acknowledgement

7.1 Author Contribution

System Implementation. Jifan Yu, Zheyuan Zhang, and Daniel Zhang-li designed the overall framework of the system. Zheyuan Zhang refined the workflow representation method used for controlling agents, while Daniel Zhang-li was responsible for the development and engineering deployment of several algorithms, including resource processing and function generation. Shangqing Tu oversaw security reviews and the integration of RAG methods. Linlu Gong collected the evaluation data from students. Nan Zhang and Ruixin Ni were responsible for project management and coordination throughout the development phase of MAIC, playing a crucial role in ensuring the quality of the final system.

Theoretical Investigation. Zhanxin Hao and Ruimiao Li were responsible for the theoretical investigation and analysis of the MAIC concept, while Yang Dang contributed to the early development of this idea. All supervising professors provided valuable insights and guidance in the conceptualization of MAIC.

Toolkit Completion. Haoxuan Li, Yuanchun Wang, Hanming Li, Jiayin Lin, Jinchang Zhou, and Nianyi Lin contributed to the development of various MAIC tools, including the cognitive diagnosis module, automated interview module, and automatic analysis module. Haohua Wang and Lijun Deng played significant roles in data collection and processing.

Course Practice. Yisi Zhan and Chaojun Xiao were instrumental in the first round of MAIC pilot courses, handling a wide range of practical tasks, including the recruitment of student volunteers, provision of course materials, content review, and post-class management. Xusheng Dai refined the

必要时可采用决策制定流程。然而，人工智能自动化与人工监督之间的平衡仍是一个复杂问题，需持续予以考量。

此外，在以人工智能为主导的教育环境中缺乏同伴互动，可能会阻碍学生重要社交技能的发展。为解决这一问题，MAIC 系统引入了模拟同伴互动的人工智能同学。虽然这些智能体能提供某种形式的互动，但无法完全复现人际交流的微妙差异。因此，该系统倡导混合模式学习环境，让学生既能与人工智能同学互动，又能与真人同学交流，从而保留社交学习的优势。

最后，学生个性化教育问题需要谨慎对待。虽然 MAIC 系统能够根据个体学习需求进行调整，这无疑是其显著优势，但也可能对教育公平性与包容性构成挑战。某些学生可能基于个人数据档案获得更精准有效的教学资源，这种现象或将加剧教育不平等现象。为有效缓解这一风险，MAIC 系统特别设计了保障机制，确保所有学生无论数据档案如何，都能获得优质学习体验。系统持续优化公平性算法，致力于为全体学习者创造均衡的教育成果。

7 结论

本文系统梳理了在线教育的发展轨迹，并深入剖析了大型语言模型时代带来的技术机遇。基于自适应性与可扩展性原则，结合LLM驱动的多智能体系统精妙设计，我们探索了如何将现有慕课（MOOC）升级为 MAIC（大规模人工智能赋能课程），并探讨了教学创新范式。我们提出综合性解决方案，解析关键技术要素，并分步实施实施流程。该方案已在清华大学两门课程中实际应用，通过学生行为数据初步验证了教学效果。研究发现表明：依托新型AI辅助学习技术构建的高度个性化课堂能够实现优质教学，学生行为数据充分印证了教学过程的有效性。未来我们将持续优化升级该平台，致力于打造开放共享的教育探索、学术研究与技术创新平台。我们希望本研究能够呼吁并服务于教育理论家、技术开发者及创新者，共同探讨大型语言模型时代下新兴的在线环境。

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7.1 作者贡献

系统实现。余继凡、张哲远与张立丹共同设计了系统整体架构。张哲远优化了用于控制智能体的工作流表示方法，张立丹负责开发并工程化部署了包括资源处理与功能生成在内的多项算法。涂尚清主导了安全审查工作及RAG方法的集成整合。龚林璐负责收集学生评估数据。张楠与倪瑞欣全程负责 MAIC 开发阶段的项目管理与协调工作，在确保最终系统质量方面发挥了关键作用。

理论研究。郝展新与李瑞淼负责 MAIC 概念的理论研究与分析，杨当则为该概念的早期发展作出贡献。所有指导教授均在 MAIC 概念化过程中提供了宝贵见解与指导。

工具包完成度。李浩轩、王远春、李汉明、林佳音、周金昌和林念义参与开发了多种 MAIC 工具，包括认知诊断模块、自动化访谈模块和自动分析模块。王浩华与邓立军在数据收集与处理方面发挥了重要作用。

课程实践。易思展与肖超军在首轮 MAIC 试点课程中发挥了关键作用，负责学生志愿者招募、课程材料提供、内容审核及课后管理等多样化实践工作。戴旭生则对课程内容进行了优化完善。

course practice process, while Xuan Yan was deeply involved in course support activities. Their efforts were critical to the successful execution of the two courses.

Pedagogical analysis. Under the guidance of Yu Zhang, Zhanxin Hao was responsible for the preliminary pedagogical evaluation of the MAIC system. Ruimiao Li, under the supervision of Manli Li, conducted key field research. Jie Cao, Fei Qin, and Jianxiao Jiang played crucial roles in the analysis process, overseeing tasks such as automated coding, qualitative interviews, and data analysis, respectively.

Paper Writing. Jifan Yu was responsible for the primary writing of the manuscript, while Zhanxin Hao and Ruimiao Li contributed significantly to the design and execution of the behavioral experiments and the ethical consideration.

Advising. Manli Li, Juanzi Li, Zhiyuan Liu, Huiqin Liu, and Maosong Sun took advisor roles in this project. Xu Han brought technical insights into the system design. This project also received guidance and support from various relevant departments at Tsinghua University.

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在课程实践过程中，宣燕深度参与了课程支持活动。他们的努力对两门课程的成功实施至关重要。

教学法分析。在余章指导下，郝展新负责 MAIC 系统的初步教学法评估工作。李瑞淼在李曼丽监督下开展关键性实地研究。曹杰、秦飞和蒋建晓在分析过程中发挥关键作用，分别负责自动化编码、定性访谈及数据分析等任务。

论文撰写。余继凡负责论文初稿撰写，郝展新与李瑞淼在行为实验设计与实施以及伦理考量方面作出重要贡献。

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